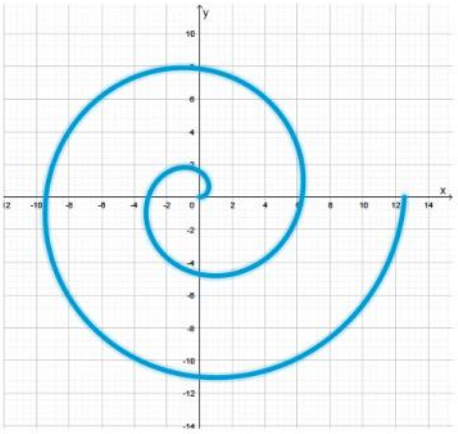
Problema 1

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Calcula la integral de línea de la función $F(x,y) = \sqrt{x^2+y^2}$ a lo largo de la espiral de parametrización $\overline{r}(t) = (t\cos(t), t\,\mathrm{sen}(t))$ para $0 \leqslant t \leqslant 4\pi$. Proporciona adicionalmente un resultado numérico con dos decimales de precisión.



$\bar{r}(t) = (t\cdot\cos(t), t\cdot\mathrm{sen}(t)), \quad t \in [0, 4\pi]$

$\bar{r}'(t) = (\cos(t) - t\,\mathrm{sen}(t), \mathrm{sen}(t) + t\cos(t))$

$dS = \|\bar{r}'(t)\| \cdot dt$

$$\int_C F(x,y)\,dS = \int_0^{4\pi} \sqrt{(t\cos(t))^2 + (t\,\mathrm{sen}(t))^2} \cdot \sqrt{(\cos(t) - t\,\mathrm{sen}(t))^2 + (\mathrm{sen}(t) + t\cos(t))}\,dt =$$

$$= \int_0^{4\pi} t \cdot \sqrt{\cos^2(t) - 2t\,\mathrm{sen}(t)\cos(t) + t^2\mathrm{sen}^2(t) + \mathrm{sen}^2(t) + 2t\,\mathrm{sen}(t)\cos(t) + t^2\cos^2(t)}\,dt =$$

$$= \int_0^{4\pi} t\sqrt{1+t^2}\,dt = \int_0^{4\pi} t(1+t^2)^{1/2}\,dt = \left[\frac{(1+t^2)^{3/2}}{3/2 \cdot 2}\right]_0^{4\pi} =$$

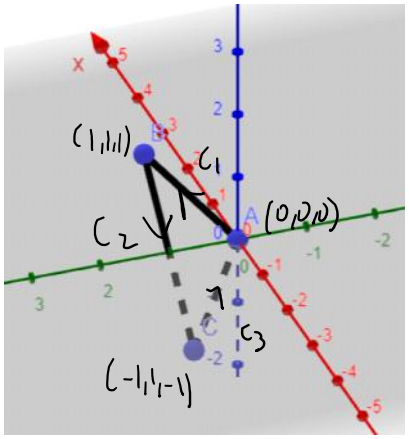
$$= \frac{1}{3}\left(\sqrt{(1+16\pi^2)^3} - 1\right) \simeq 667'43$$

Problema 2

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Comprueba que se cumple el teorema de Stokes al calcular el trabajo realizado por el campo $\overline{F}(x,y,z) = (yz, xz, x(y+1))$ para desplazar una partícula que recorra el triángulo de vértices $(0,0,0)$, $(1,1,1)$ y $(-1,1,-1)$ en ese orden.

1) Cálculo de la integral de línea: $\text{Trabajo} = \int_{C=C_1 \cup C_2 \cup C_3} \overline{F}(\bar{r}(t))\,\bar{r}'(t)\,dt$



C_1: $\bar{r}(t) = (t, t, t)$, $t_a = 0$, $t_b = 1$

$\bar{r}'(t) = (1, 1, 1)$

C_2: $\bar{r}(t) = (1-t, 1, 1-t)$, $t_a = 0$, $t_b = 2$

$\bar{r}'(t) = (-1, 0, -1)$

C_3: $\bar{r}(t) = (t-1, 1-t, t-1)$, $t_a = 0$, $t_b = 1$

$\bar{r}'(t) = (1, -1, 1)$

$$\text{Trabajo} = \int_0^1 \left(t\cdot t,\ t\cdot t,\ t(t+1)\right)\cdot(1,1,1)\,dt + \int_0^2 \left(1\cdot(1-t),\ (1-t)^2,\ (1-t)\cdot 2\right)\cdot(-1,0,-1)\,dt +$$

$$+ \int_0^1 \left(\underbrace{(1-t)(t-1)}_{-t^2+2t-1},\ \underbrace{(t-1)^2}_{t^2-2t+1},\ \underbrace{(t-1)(1-t+1)}_{-t^2+3t-2}\right)(1,-1,1)\,dt$$

$$= \int_0^1 (t^2+t^2+t^2+t)\,dt + \int_0^2 (t-1+2t-2)\,dt + \int_0^1 \left(-(t-1)^2-(t-1)^2-(t-1)^2+t-1\right)dt =$$

$$= \int_0^1 (3t^2+t)\,dt + \int_0^2 (3t-3)\,dt + \int_0^1 (-3t^2+7t-4)\,dt =$$

$$= \left[t^3+\frac{t^2}{2}\right]_0^1 + \left[\frac{3t^2}{2}-3t\right]_0^2 + \left[-t^3+\frac{7t^2}{2}-4t\right]_0^1 =$$

$$= \left(1+\frac{1}{2}\right) + \left(6-6\right) + \left(-1+\frac{7}{2}-4\right) = \boxed{0}$$

2) Cálculo de la integral de superficie

$$\nabla \times \overline{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xz & x(y+1) \end{vmatrix} = (x-x,\ y-(y+1),\ z-z) = (0,-1,0)$$

El plano que contiene el triángulo es $x - z = 0 \Rightarrow$

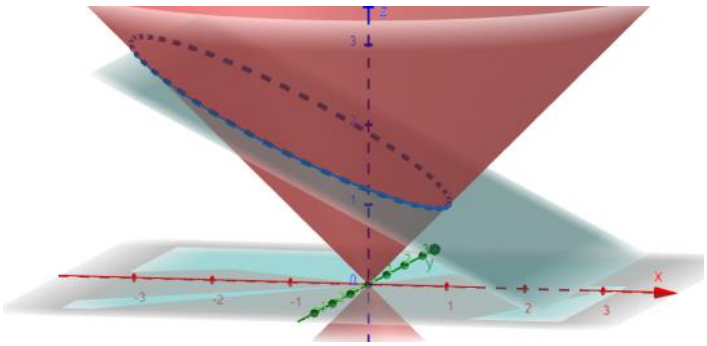
$\Rightarrow \bar{n} = (1, 0, -1) \Rightarrow \bar{N} = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right)$

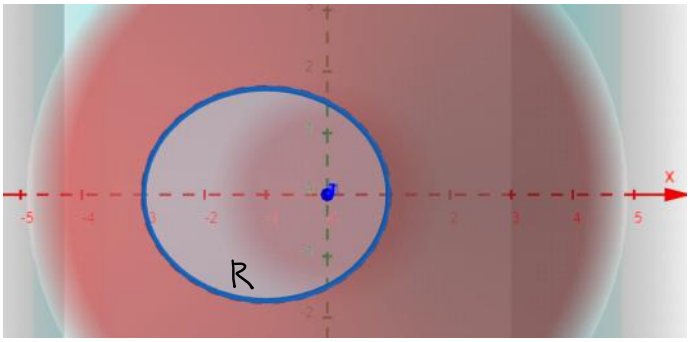
$$\oint_C \bar{F} \cdot d\bar{r} = \iint_S (\nabla \times \bar{F}) \bar{N} \, dS = \iint_S (0, -1, 0) \cdot \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right) dS = \boxed{0}$$

Problema 3

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Calcula el área lateral de la porción del cono $z^2 = x^2 + y^2$ limitada entre los planos $x + 2z = 3$ y $z = 0$.





El área de una superficie se calcula como $\iint_S dS$. Vamos a proyectar sobre el plano XY. Vamos a determinar el recinto R.

$$\left.\begin{array}{l} z^2 = x^2+y^2 \\ z = \frac{1}{2}(3-x) \end{array}\right] \Rightarrow \frac{1}{4}(3-x)^2 = x^2+y^2 \Rightarrow 9-6x+x^2 = 4x^2+4y^2 \Rightarrow$$

$$\Rightarrow 3x^2+6x+4y^2 = 9 \Rightarrow 3(x+1)^2+4y^2 = 12 \Rightarrow \left(\frac{x+1}{2}\right)^2+\left(\frac{y}{\sqrt{3}}\right)^2 = 1$$

$$z^2 = x^2+y^2 \longrightarrow z = \pm\sqrt{x^2+y^2} \xrightarrow{z>0} z = f(x,y) = \sqrt{x^2+y^2}$$

$$f(x,y) = (x^2+y^2)^{1/2} \Rightarrow f_x(x,y) = \frac{x}{\sqrt{x^2+y^2}}, \quad f_y(x,y) = \frac{y}{\sqrt{x^2+y^2}}$$

$$\iint_S dS = \iint_R \sqrt{1+(f_x(x,y))^2+(f_y(x,y))^2}\,dx\,dy = \iint_R \sqrt{1+\left(\frac{x}{\sqrt{x^2+y^2}}\right)^2+\left(\frac{y}{\sqrt{x^2+y^2}}\right)^2}\,dx\,dy =$$

$$= \iint_R \sqrt{1+\frac{x^2}{x^2+y^2}+\frac{y^2}{x^2+y^2}}\,dx\,dy = \iint_R \sqrt{2}\,dx\,dy =$$

$$= \left\{\begin{array}{l} \frac{x+1}{2} = r\cdot\cos(\theta) \longrightarrow x = -1+2\cdot r\cdot\cos(\theta) \\ \frac{y}{\sqrt{3}} = r\cdot\operatorname{sen}(\theta) \longrightarrow y = \sqrt{3}\cdot r\cdot\operatorname{sen}(\theta) \end{array}\right. \quad |J| = \begin{vmatrix} 2\cos(\theta) & -2r\operatorname{sen}(\theta) \\ \sqrt{3}\operatorname{sen}(\theta) & \sqrt{3}r\cos(\theta) \end{vmatrix} = 2\sqrt{3}\cdot r \Big\{ =$$

$$= \int_{\theta=0}^{\theta=2\pi} \left(\int_{r=0}^{r=1} \sqrt{2}\cdot 2\sqrt{3}\ \ r\cdot dr \right) d\theta = 2\sqrt{6} \left(\int_{\theta=0}^{\theta=2\pi} d\theta \right) \cdot \left(\int_{r=0}^{r=1} r\cdot dr \right) =$$

$$= 2\sqrt{6}\cdot [\theta]_0^{2\pi} \cdot \left[\frac{r^2}{2} \right]_0^1 = 2\sqrt{6}\ 2\pi \cdot \frac{1}{2} = \boxed{2\cdot\sqrt{6}\cdot\pi}$$

Problema 4

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Calcula el flujo del campo vectorial $\overline{F} = (xy^2 + z^2, x^2y + z^2, z(x^2 + y^2))$ hacia el exterior del cubo limitado por los planos $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$.

Vamos a aplicar el teorema de Gauss: $\iiint_D (\nabla \cdot \bar{F})\, dV = \iint_S \bar{F} \cdot \bar{N} \cdot dS$

$$\nabla \cdot \bar{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot \left(xy^2 + z^2,\ x^2y + z^2,\ z(x^2+y^2) \right) =$$

$$= y^2 + x^2 + (x^2+y^2) = 2(x^2+y^2)$$

$$\iint_S \bar{F} \cdot \bar{N} \cdot dS = \iiint_D (\nabla \cdot \bar{F})\, dV = \int_{x=0}^{x=1} \left(\int_{y=0}^{y=1} \left(\int_{z=0}^{z=1} 2(x^2+y^2)\, dz \right) dy \right) dx =$$

$$= 2 \int_{x=0}^{x=1} \left(\int_{y=0}^{y=1} (x^2+y^2) \cdot [z]_0^1\, dy \right) dx = 2 \int_{x=0}^{x=1} \left(\int_{y=0}^{y=1} (x^2+y^2)\, dy \right) dx =$$

$$= 2 \int_{x=0}^{x=1} \left(x^2 [y]_0^1 + \left[\frac{y^3}{3} \right]_0^1 \right) dx = 2 \int_{x=0}^{x=1} \left(x^2 + \frac{1}{3} \right) dx = 2 \left[\frac{x^3}{3} + \frac{x}{3} \right]_0^1 = \boxed{\frac{4}{3}}$$